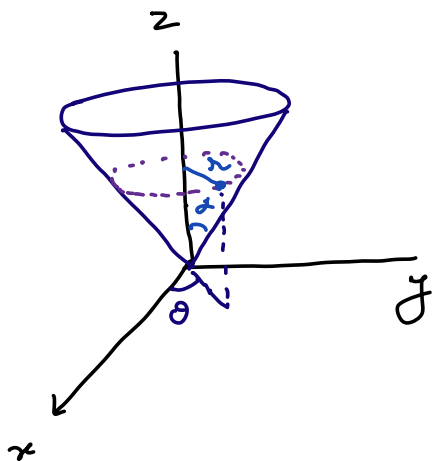


Solutions to Tutorial 5

(1)



Let us choose r , θ and z as the generalized coordinates.

But, we have the equation of constraint

$$z = r \cot \alpha \rightarrow (1)$$

So, there are just two degrees of freedom
 $\Rightarrow$ there are only two proper generalized coordinates. We may use (1) to eliminate either z or r ; we choose to eliminate z .

Now,

$$\begin{aligned} v^2 &= \dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2 \\ &= \dot{r}^2 + r^2 \dot{\theta}^2 + \dot{r}^2 \cot^2 \alpha \\ &= \dot{r}^2 \operatorname{cosec}^2 \alpha + r^2 \dot{\theta}^2 \end{aligned}$$

The potential energy is

$$V = mgz = mgr \cot \alpha$$

(we choose $V=0$ at $z=0$)

Thus, the Lagrangian is

$$L = \frac{1}{2} m (\dot{r}^2 \csc^2 \alpha + r^2 \dot{\theta}^2) - mg r \cot \alpha$$

The Lagrange's equations for θ and r are:

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = 0 \Rightarrow 0 - \frac{d}{dt} (mr^2 \dot{\theta}) = 0$$

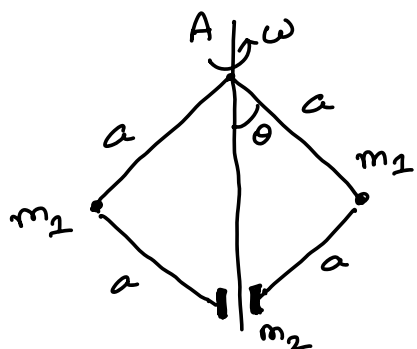
$$\Rightarrow \boxed{mr^2 \dot{\theta} = \text{constant}}$$

$$\frac{\partial L}{\partial r} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) = 0$$

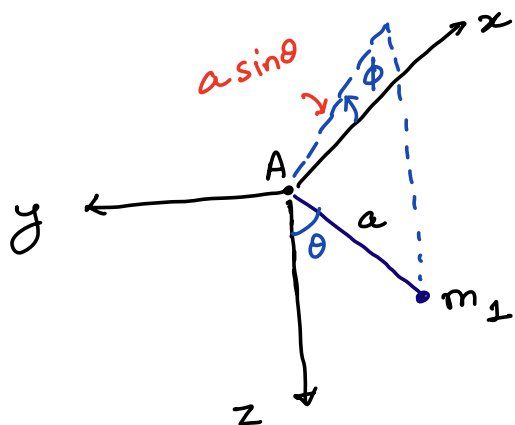
$\Rightarrow$

$$\boxed{\ddot{r} - r \dot{\theta}^2 \sin^2 \alpha + g \sin \alpha \cos \alpha = 0}$$

(2)



Let us first look at one of the masses, m_1 .



Here $\dot{\phi} = \omega$

For each particle m_1 , the infinitesimal displacement is given by

$$d\mathbf{r}_1^2 = a^2 d\theta^2 + a^2 \sin^2 \theta d\phi^2$$

Note: $a d\theta$ is the length element along $\hat{\theta}$
and $a \sin \theta d\phi$ is the length element along $\hat{\phi}$

The distance of m_2 from the support A is

$$2a \cos \theta$$

So, $dl_2 = -2a \sin \theta d\theta$

Thus,

$$\begin{aligned} T &= \frac{1}{2} m_1 \dot{l}_1^2 + \frac{1}{2} m_2 \dot{l}_2^2 + \frac{1}{2} m_1 \dot{l}_1^2 \\ &= m_1 (a^2 \dot{\theta}^2 + a^2 \sin^2 \theta \dot{\phi}^2) \\ &\quad + \frac{1}{2} m_2 (4a^2 \sin^2 \theta \dot{\theta}^2) \\ &= m_1 a^2 (\dot{\theta}^2 + \omega^2 \sin^2 \theta) \\ &\quad + 2 m_2 a^2 \sin^2 \theta \dot{\theta}^2 \end{aligned}$$

$$\begin{aligned} V &= -2m_1 g a \cos \theta - 2m_2 g a \cos \theta \\ &= -2(m_1 + m_2) g a \cos \theta \end{aligned}$$

Hence,

$$\begin{aligned} L &= m_1 a^2 (\dot{\theta}^2 + \omega^2 \sin^2 \theta) + 2m_2 a^2 \dot{\theta}^2 \sin^2 \theta \\ &\quad + 2(m_1 + m_2) g a \cos \theta \end{aligned}$$

//

(3)

Because of cylindrical symmetry
we choose r, θ and z as the coordinates

$$v^2 = \dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2$$

But $z = cr^2, \quad \theta = \omega t$

So, $\dot{z} = 2cr\dot{r}, \quad \dot{\theta} = \omega$

The kinetic energy of the bead is

$$\begin{aligned} T &= \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + \dot{z}^2) \\ &= \frac{1}{2} m (\dot{r}^2 + r^2 \omega^2 + 4c^2 r^2 \dot{r}^2) \end{aligned}$$

The potential energy is

$$\begin{aligned} V &= mgz \\ &= mgr^2 \end{aligned} \quad \left(\begin{array}{l} \text{we choose } V=0 \\ \text{at } z=0 \end{array} \right)$$

Hence,

$$L = T - V$$

$$= \frac{1}{2} m (\dot{r}^2 + r^2 \omega^2 + 4c^2 r^2 \dot{r}^2) - mgr^2$$

Lagrange's equation of motion

$$\frac{\partial L}{\partial r} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) = 0 \quad \text{results in}$$

$$\ddot{r} (1 + 4c^2 r^2) + \dot{r}^2 (4c^2 r) + r (2gc - \omega^2) = 0 \quad \rightarrow (1)$$

Now, it is given that the bead rotates with $r = R = \text{constant}$.

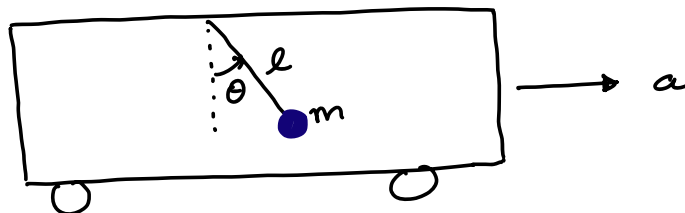
So, $\dot{r} = 0 = \ddot{r}$.

Eq. (1) becomes:

$$R (2gc - \omega^2) = 0$$

$$\Rightarrow \boxed{c = \frac{\omega^2}{2g}} //$$

(4)



The pendulum has mass m , length l and displacement angle θ .

Let us choose a fixed cartesian coordinate system with $x=0$ and $\dot{x}=v_0$ at $t=0$.

The position and velocity of m :

$$x = v_0 t + \frac{1}{2} a t^2 + l \sin \theta$$

$$y = -l \cos \theta$$

$$\dot{x} = v_0 + a t + l \dot{\theta} \cos \theta$$

$$\dot{y} = l \dot{\theta} \sin \theta$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) ; \quad V = -mg l \cos \theta$$

The Lagrangian is

$$\begin{aligned} L &= T - V \\ &= \frac{1}{2} m (v_0 + a t + l \dot{\theta} \cos \theta)^2 + \frac{1}{2} m l^2 \dot{\theta}^2 \sin^2 \theta \\ &\quad + mg l \cos \theta \end{aligned}$$

Clearly the angle θ is the only generalized coordinates.

Lagrange's equation

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = 0 \quad \text{results in}$$

$$\boxed{\ddot{\theta} = -\frac{g}{l} \sin \theta - \frac{a}{l} \cos \theta} \rightarrow (1)$$

We can determine the equilibrium angle $\theta = \theta_e$ by setting $\ddot{\theta} = 0$.

$$g \sin \theta_e + a \cos \theta_e = 0$$
$$\Rightarrow \tan \theta_e = -\frac{a}{g} \rightarrow (2)$$

Because oscillations are small and are about the equilibrium angle, let us take:

$$\theta = \theta_e + \eta, \text{ where } \eta \text{ is a small angle.}$$

From (1):

$$\ddot{\theta} = \ddot{\eta} = -\frac{g}{l} \sin(\theta_e + \eta) - \frac{a}{l} \cos(\theta_e + \eta)$$

$$\Rightarrow \ddot{\eta} = -\frac{g}{l} \left[\sin \theta_e \cos \eta + \cos \theta_e \sin \eta \right] - \frac{a}{l} \left[\cos \theta_e \cos \eta - \sin \theta_e \sin \eta \right]$$

Under small angle approximation we can take:

$$\cos \eta \approx 1, \sin \eta \approx \eta$$

$$\left[\begin{array}{l} \text{recall} \\ \sin 0^\circ = 0 \\ \cos 0^\circ = 1 \end{array} \right]$$

Thus under small angle approximation we get:

$$\boxed{\ddot{\eta} = -\frac{1}{l} (g \cos \theta_e - a \sin \theta_e) \eta} \rightarrow (3)$$

$$(\text{noting, } g \sin \theta_e + a \cos \theta_e = 0)$$

Now, $\tan \theta_e = -\frac{a}{g}$

So, $\sin \theta_e = \frac{a}{\sqrt{g^2 + a^2}}$, $\cos \theta_e = \frac{g}{\sqrt{g^2 + a^2}}$

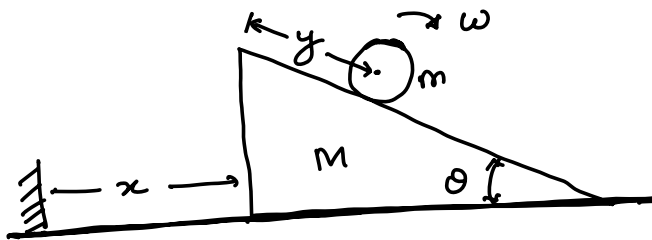
Hence, from (3) we obtain:

$$\ddot{\eta} = -\frac{\sqrt{a^2 + g^2}}{l} \eta$$

This equation represents a simple harmonic motion, with frequency ω

given by $\omega^2 = \frac{\sqrt{a^2 + g^2}}{l}$

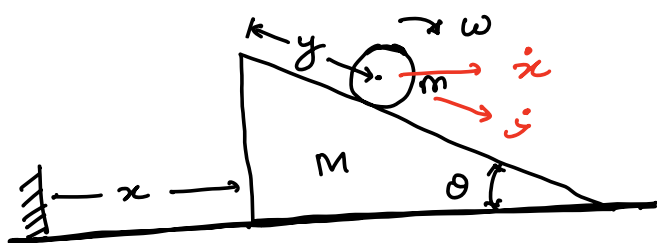
(5)



Applying the rolling condition, the angular velocity ω of the ball is given by:

$\omega = \dot{y}/a$, a is the radius of the ball

Also, $I = \frac{2}{5} m a^2$, I is the moment of inertia of the ball.



The square of the velocity of the ball is

$$v^2 = \dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\cos\theta$$

The kinetic energy of the system is

$$T = \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\cos\theta) + \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\cos\theta) + \frac{1}{2} \left(\frac{2}{5} m a^2 \right) \left(\frac{\dot{y}}{a} \right)^2$$

$$\Rightarrow T = \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m \left(\dot{x}^2 + \frac{7}{5} \dot{y}^2 + 2\dot{x}\dot{y}\cos\theta \right)$$

The potential energy of the system

$$V = -mg y \sin\theta, \quad \text{for } V=0 \text{ at the initial position of the ball.}$$

Thus,

$$L = \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m \left(\dot{x}^2 + \frac{7}{5} \dot{y}^2 + 2\dot{x}\dot{y}\cos\theta \right) + mg y \sin\theta$$

Lagrange's eq^{ns}

$$\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = 0$$

$$\frac{\partial L}{\partial y} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}} \right) = 0$$

leads to :

$$\ddot{x} = \frac{5mg \sin \theta \cos \theta}{5m \cos^2 \theta - 7(m+M)}$$

↑ This is the required acceleration of the wedge.